Deliverable 2: Cryptographic Implementation Plan (Task 2)

1. Objective

To implement a robust cryptographic framework using industry-standard asymmetric encryption and hashing to guarantee the authenticity and integrity of all IoT firmware updates.

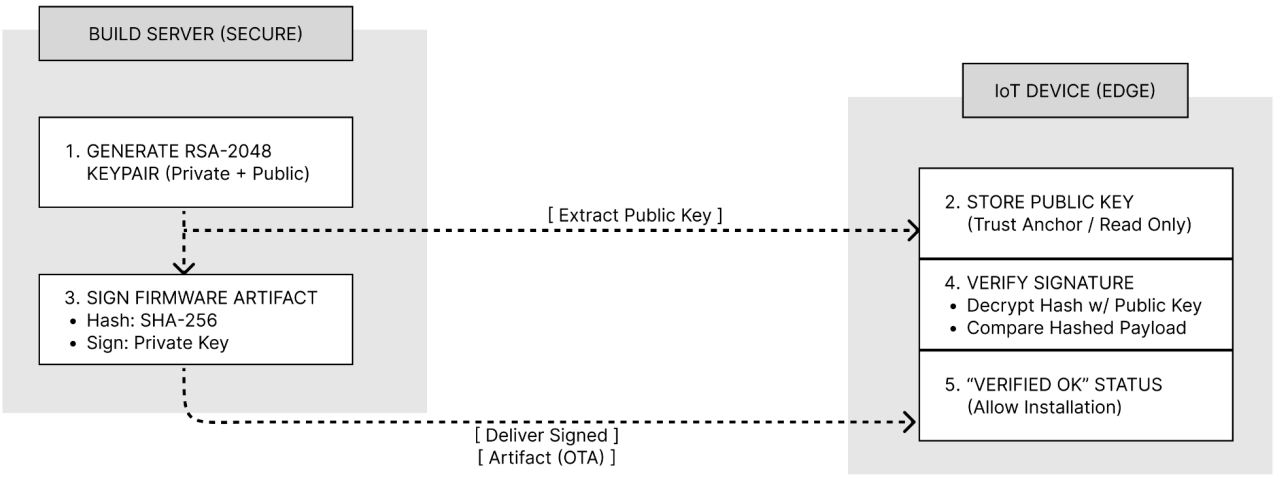
2. Cryptographic Specifications

This section outlines the specific algorithms and parameters selected for the project.

CRYPTOGRAPHIC STANDARDS	
FEATURE	SPECIFICATION
Asymmetric Algorithm	RSA-2048 (Rivest-Shamir-Adleman)
Hashing Algorithm	SHA-256 (Secure Hash Algorithm)
Key Format	PKC#1 (Traditional RSA/Legacy)
Encoding	X.509 PEM (Private Enhanced Mail)

3. Key Management Lifecycle

This diagram illustrates how keys are generated and utilized across the build server and the target device.



4. Implementation Validation

- **Key Conversion:** To ensure compatibility with the Mender-Artifact tool, private keys were explicitly forced into the **PKCS#1** format using the `-traditional` flag in OpenSSL.
- **Signature Verification:** The integrity of the payload was manually validated using the public key, resulting in a successful verification output.
    - **Result:** `Verified OK`